AI-use disclosure (outside the 3-page report limit)

Tool used. Claude (Anthropic; Claude Code agent, model Claude Opus 5.5), during the hackathon window.

What the AI did. - Read the problem statement, `scorer.py` and `scenario_R.json`, and proposed the modelling approach (per-approach decomposition, quadratic Walsh projection, adaptive measure) and the QAOA design (multi-angle mixer, split cost angles, Gibbs objective, linear-ramp initialisation, two-stage budgeted COBYLA). - Wrote the code in this repository (`run.py`, `traffic_model.py`, `qubo.py`, `qaoa.py`, `circuit.py`, `baselines.py`, `experiments/`, `tests/`) and drafted `README.md` and `REPORT.md`. - In a second round, ran five parallel research sub-agents (analytic QUBO, optimiser and ansatz, transfer initialisation, three-intersection chain, shot-based training and hardware) under the same rules. Each wrote only under `lab/<track>/` and reported against a common validation set (R + seed 555). Two tracks were also re-run by independent reviewer agents. The adopted changes (`analytic_qubo.py`, `qaoa_nft.py`) were then re-checked by the repository tests and re-benchmarked on held-out sets. - Ran the exploratory sweeps used to choose hyper-parameters. They were selected on randomly generated scenarios (generator seeds 1, 7, 11, 31) and then **frozen**. The reported generalization (seed 2026) and ablation (seed 99) sets were not used for selection. Scenario R was never used to tune anything beyond being one point among many.

How results were independently verified. - `scorer.py` is the unmodified official file. All reported costs come from `scorer.evaluate`. The four published verification vectors are re-checked in `tests/test_pipeline.py`. - The per-approach decomposition is checked against the scorer on 1,240 random (scenario, schedule) pairs. - The QUBO↔Ising conversion, the Walsh-orthogonality of the uniform projection, and the identity between the numpy trainer and the Qiskit circuit (max probability difference ≈ 1e-15) are asserted on every run. - OpenQASM 3 round trip and counts-vs-circuit consistency are tested. Counts come from Qiskit's seeded `StatevectorSampler` on the exported circuit. - Budget compliance (≤ 300 evaluations, $p \le 4$, ≤ 16 qubits, exactly 4,096 shots) is asserted at run time. - One AI-introduced bug was caught by these checks during development: an inverted mixer sign that made the annealing initialisation start on the wrong side. It was fixed before any reported run.

What the AI did not do. It did not access hidden scenarios. It did not hard-code any schedule or scenario-specific value, and it did not use the official cost of any schedule inside the QAOA objective. The decoded raw-QAOA schedule is chosen by QUBO energy only.

Team responsibility. The team reviewed the derivation, reran every command in `README.md`, and takes responsibility for the submission. *(Team: please edit this paragraph to describe your own contributions accurately.)*